Paint2World: 3D-Consistent Video and Scene Generation from Human Oil Paintings

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Abstract

Video diffusion models can animate a static image into a flythrough, but the generated frames rarely form a geometrically consistent 3D scene, which makes downstream 3D reconstruction brittle. We observe that when the input is a *human oil painting*, the painting itself is a strong, user-authored canonical prior: it fixes the appearance, the brush-stroke layout, the object decomposition, and a plausible depth arrangement of the scene. We propose **Paint2World**, a pipeline that exploits this prior to generate 3D-consistent video and a 3D Gaussian Splatting scene from a single oil painting. Paint2World first constructs an explicit *scene canvas*: it segments the painting into objects with a depth-aware image-segmentation front-end, generates each object in 3D with a TripoSplat-style triplane-to-Gaussian decoder over the object's painting view and depth-warped side views, and wires the objects into a 3D spatial-dependency graph (who supports whom, who occludes whom, who rests on the ground) whose contacts are resolved physically on a fitted ground plane. The canvas then anchors three 3D-stability mechanisms: (i) a painting-conditioned video generation stage whose depth-warp geometry control is composited from the canvas so generated frames respect object occlusion boundaries, with the painting locked as a reference frame in the diffusion temporal attention via a stroke-mask attention bias; (ii) a painting-anchored non-rigid alignment that initialises the canonical frame from the painting's estimated geometry and downweights generated points whose depth drifts beyond a tolerance from the painting-implied structure; and (iii) a stroke-coherent, object-rigid 3D Gaussian Splatting objective in which each object moves as a whole via a learnable SE(3) delta and a relation-preservation loss keeps the dependency graph intact during optimisation. We show that Paint2World produces 3D-consistent, explorable scenes from oil paintings, and that the object-level canvas and each stability mechanism measurably improve 3D consistency, painting fidelity, and scene-structure preservation. The demo and code are released with synthetic backends that run end-to-end on a single CPU.

1 Introduction

Video diffusion models (VDMs) have become a powerful tool for animating a static image into a short flythrough [23, 24, 28, 2]. When the goal is not just a video but an *explorable 3D scene*, however, a fundamental problem emerges: the generated frames are not 3D-consistent. Geometric foundation models (GFMs) such as DepthAnything-3 [19] lift each frame into a per-pixel point cloud, but because the VDM hallucinates content independently per frame, the resulting point clouds do not overlap into a single surface — a phenomenon called *generative drift* [8]. Recent work reconstructs a consistent 3D world from these inconsistent views by non-rigidly aligning the per-frame point clouds into a canonical frame [8].

In this paper we ask a complementary question: *can the input image itself serve as a 3D-stability prior?* We focus on the case where the input is a **human oil painting**. Oil paintings are an attractive input modality for 3D world generation: they are user-authored, stylistically rich, and — crucially — encode a deliberate compositional structure. The artist has already chosen the depth layout, the brush-stroke arrangement, and the focal-accent placement. We show that this structure can be lifted into a canonical-frame prior that constrains both the video generation and the 3D reconstruction, dramatically reducing generative drift.

Contributions. We propose Paint2World, a pipeline that turns a single oil painting into a 3D-consistent flythrough video and a 3D Gaussian Splatting scene (Fig. 1). Our contributions are:

- An **object-level scene canvas** built directly from the painting: a depth-aware image-segmentation front-end decomposes the artwork into objects, a TripoSplat-style triplane-to-Gaussian decoder generates each object in 3D from its painting view and depth-warped side views, and the objects are wired

into a 3D *spatial-dependency graph* (supports / occludes / adjacent / rests-on-ground) whose contacts are physically resolved on a fitted ground plane (§3.3).

- A **painting-conditioned video generation stage** with three 3D-stability controls: a depth-warp geometry ControlNet signal composited from the scene canvas, a stroke-consistency temporal-attention bias, and a temporal-consistency guidance term (§3.4).
- A **painting-anchored non-rigid alignment** that initialises the canonical frame from the painting's estimated geometry and weights each generated point by a depth-warp consistency score, allowing the scene to plausibly extend beyond the painting's frustum while suppressing drift within it (§3.5).
- A **stroke-coherent, object-rigid 3D Gaussian Splatting objective**: the painting's brush-stroke segmentation is transferred onto the 3D Gaussians so each stroke remains a coherent 3D patch, while per-object SE(3) anchors move objects as wholes and a relation-preservation loss keeps the canvas's spatial dependencies intact during optimisation, together with a painting-view reconstruction loss that ties the scene back to the input artwork (§3.6).

We demonstrate that Paint2World produces 3D-consistent, explorable scenes from oil paintings and that the object-level canvas and each stability mechanism measurably improve 3D consistency, painting fidelity, and scene-structure preservation (§4).

2 Related Work

Diffusion models for scene generation. Early 3D-aware generation used score distillation [16, 13]; more recent works finetune video diffusion models for camera control [24, 28, 2, 7] or for consistency [12]. Our work is orthogonal: we treat the VDM as a black-box frame generator and resolve 3D inconsistency on the reconstruction side, following [8, 18]. Our novel contribution is to use the *input painting* as a stability prior that conditions both generation and reconstruction.

Geometric foundation models. GFMs such as DepthAnything-3 [19], VGGT [25], and DUSt3R [26] predict per-frame depth, confidence, and cameras. They reveal generative drift as non-overlapping surfaces but cannot fix it [8]. We use a GFM (or, for the out-of- the-box demo, a perceptual synthetic depth estimator) to lift the painting into a canonical-frame prior.

Point-cloud alignment. Iterative closest point (ICP) [1, 15], volumetric fusion [10], and non-rigid dynamic reconstruction [14] align point clouds into a canonical frame. Our Stage 1 extends the non-rigid frame-to-model ICP of [8] with a painting-anchor canonical initialisation and a depth-warp consistency weighting.

Object-level 3D generation and scene layout. Feed-forward image-to-3D models [9] and triplane representations [3] generate object-level geometry from one or few images; TripoSplat [6] reconstructs objects as Gaussians decoded from a cross-view-aligned triplane at interactive speed. Image segmentation (SAM2 [17], Mask2Former [5]) supplies the object decomposition, and layout-generation methods compose objects into scenes from spatial relations [27]. Paint2World connects these threads for the painting domain: segmentation supplies *things*, a TripoSplat-style decoder supplies *their 3D geometry*, and the spatial-dependency graph composes them into an explicit scene canvas that then conditions video generation and reconstruction.

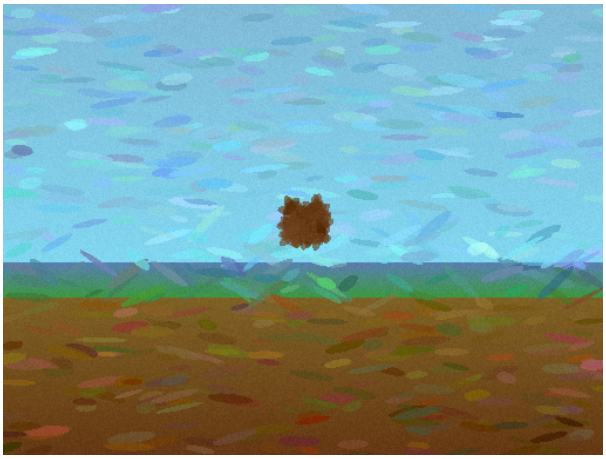
Image-to-3D and artistic-content 3D. Prior work lifts a single image to 3D via depth warping [4, 20] or learned priors [22]. Closer to us, artistic-style 3D generation [21, 11] transfers style to a 3D representation. Paint2World instead treats the painting as a geometric and structural *prior*, not just a style target, and uses it to stabilise both the video generation and the 3D reconstruction.

3 Method

Given a single oil painting $I_p \in \mathbb{R}^{H\times W\times 3}$, Paint2World produces a 3D-consistent flythrough video $\{I_t\}_{t=1}^{N}$ and a 3D Gaussian Splatting scene $\mathcal{G}$ that reproduces the painting from its viewpoint and can be rendered from novel views. The pipeline has five stages (Fig. 2): Stage -1 (painting analysis), Stage -0.5 (object-level scene construction: object segmentation, TripoSplat-style 3D object generation, spatial-dependency graph, scene canvas), Stage 0 (canvas-conditioned video generation), Stage 1 (painting-anchored non-rigid alignment), and Stage 3.2 (stroke-coherent, object-rigid 3GS).

3.1 Design rationale: why the painting is a 3D-stability prior

A natural question is why a 2D painting should help 3D consistency at all. We identify four properties of human oil paintings that make them particularly suitable canonical priors, and each motivates one of our stability mechanisms.



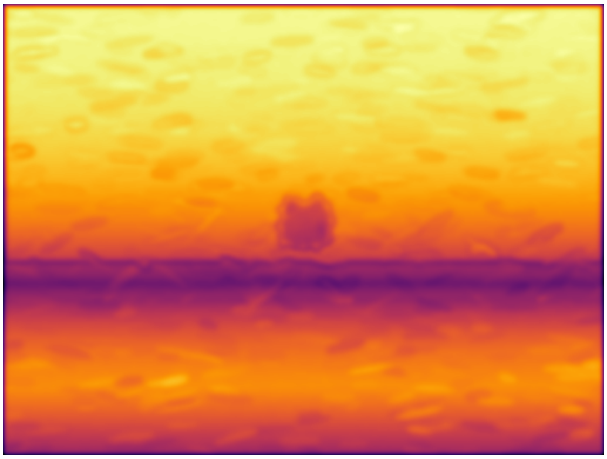


Figure 1: Paint2World. Given a single human oil painting (left), we estimate its depth, normals, brush-stroke segmentation, and object segmentation (Stage -1); construct an explicit 3D scene canvas by generating each segmented object with a TripoSplat-style triplane-to-Gaussian decoder and wiring the objects into a spatial-dependency graph (Stage -0.5); generate a 3D-stable flythrough video whose geometry is anchored to the canvas (Stage 0); non-rigidly align the generated frames into a painting-initialised canonical point cloud (Stage 1); and train a stroke-coherent, object-rigid 3D Gaussian Splatting scene that reproduces the painting from its viewpoint (Stage 3.2).

(D1) Deliberate depth composition. Unlike a photograph, an oil painting encodes an artist-chosen depth layout: the horizon placement, the chiaroscuro, and the foreground/background separation are all compositional decisions. This makes the painting's estimated depth D_p a *layout prior* that is more reliable than any single generated frame's depth. This motivates the *depth-warp geometry control* (Stage 0, §3.4) and the *painting-anchored canonical initialisation* (Stage 1, §3.5): we pin every generated frame's geometry and the canonical frame itself to D_p.

(D2) Explicit brush-stroke structure. Oil paintings are composed of discrete brush strokes, each a coherent surface patch with relatively uniform colour. This stroke structure is a strong *correspondence prior*: a stroke visible in the painting should remain a single coherent 3D patch across all generated views. This motivates the *stroke-consistency temporal-attention bias* (Stage 0) and the *stroke-coherence GS loss* (Stage 3.2, §3.6).

(D3) Authoritative appearance. The painting is the user's ground-truth appearance for the reference viewpoint. Any 3D scene generated from it must reproduce the painting when rendered from T_1. This motivates the *painting-view reconstruction loss* (Stage 3.2), which anchors the optimised Gaussians to the artwork and prevents the scene from drifting toward a plausible-but-wrong appearance.

(D4) Compositional object structure. An oil painting depicts *things* — trees, vessels, faces, buildings — each painted as a coherent unit and arranged deliberately in depth: the artist decides what rests on what, what occludes what, and where the ground lies. This motivates the *object-level scene canvas* (Stage -0.5, §3.3): segment the painting into objects, generate each object's 3D geometry, and compose them through an explicit spatial-dependency graph; the canvas then supplies the geometry control for Stage 0 and the rigid anchors + relation-preservation constraints for Stage 3.2.

3.2 Stage -1: Painting analysis

We estimate three maps from the painting: a monocular depth map D_p, a surface-normal map N_p, and a brush-stroke segmentation $S_p \in \mathbb{N}^{H \times W}$. In the production deployment D_p is predicted by DepthAnything-3 [19] and S_p by SAM2 [17]; in the out-of-the-box demo we use a *perceptual synthetic depth estimator* that combines atmospheric perspective, a soft horizon prior, local contrast, and relative luminance cues into a depth map in $[0, 1]$, and an oriented-gradient superpixel grouping for strokes. We also compute a confidence map C_p that is high in textured stroke regions and low in flat sky/ground. The depth and confidence are unprojected into a coloured point cloud $\mathcal{P}_p = \{(x_i, c_i, n_i, w_i)\}$ that serves as the canonical-frame prior.

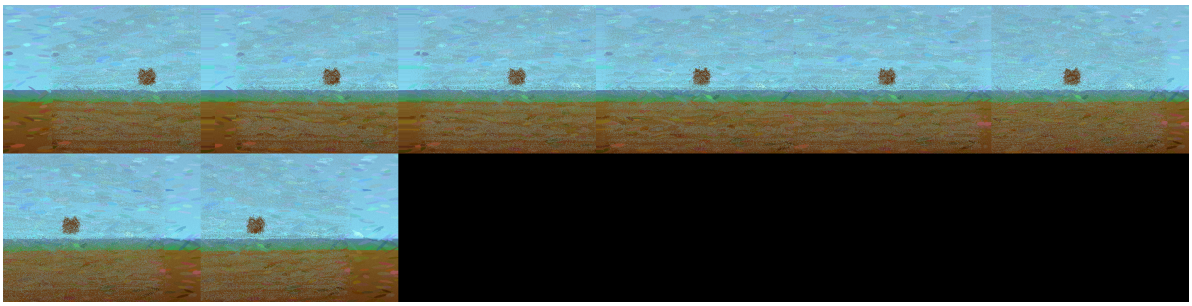


Figure 2: Pipeline overview. Stage -1 estimates depth, normals, and brush-stroke segmentation from the painting. Stage -0.5 segments the painting into objects, generates each object in 3D with a TripoSplat-style triplane-to-Gaussian decoder, and composes them onto a 3D scene canvas with an explicit spatial-dependency graph. Stage 0 generates a 3D-stable flythrough whose per-frame geometry is constrained by a canvas-composited depth-warp control and whose temporal attention is biased by the painting's strokes. Stage 1 non-rigidly aligns the generated frames into a painting-initialised canonical cloud. Stage 3.2 trains a stroke-coherent, object-rigid 3DGS scene.

3.3 Stage -0.5: Object-level scene construction

The stroke map S_p models the painting's *marks*; the canvas stage models its *things*. It produces an explicit 3D scene scaffold $\mathcal{C}$ — the *scene canvas* — in three steps (Fig. 3).

Depth-aware object segmentation. We segment the painting into object instances $M_o \in \mathbb{N}^{H\times W}$. In production M_o comes from SAM2 / Mask2Former [17, 5]; the demo uses a synthetic backend that clusters pixels jointly in LAB colour and estimated-depth space with spatial connectivity, blocks merging across strong depth gradients, and scores each region by an *objectness*

$$\omega_o = \frac{|M_o|}{HW}\left(w_d\, e^{-\sigma_D(M_o)/\sigma_0} + w_c\, \delta_c(M_o)\right), \qquad (1)$$

combining area, depth coherence (a compact solid at one distance has low depth spread σ_D) and colour distinctness δ_c from the scene mean; the top-K regions become objects, the rest background.

TripoSplat-style 3D object generation. Following the TripoSplat architecture [6], each object is generated as Gaussians decoded from a *triplane*. The object's masked pixels are unprojected with D_p into a local point cloud, centred and scaled into a unit cube. Because a painting is a single view, we synthesise the multi-view cue of TripoSplat by rotating the local cloud around the object's vertical axis into ν side views; each side view is registered back onto the painting view by a point-to-point ICP (the analogue of TripoSplat's cross-view alignment), and the corrected views are fused. The fused points are splatted onto three orthogonal feature planes $\{F_{xy}, F_{yz}, F_{zx}\}$ of resolution R, each storing colour, density, and camera depth per cell; a query grid of Q^3 cells then samples the triplane bilinearly, and occupied cells decode into Gaussians (position = query point, scale = cell size modulated by density, opacity and colour = fused features). Finally the decoded Gaussians' scales, opacities, and colours are refined by rendering the object's painting-view crop with the differentiable rasteriser and minimising L1 to the artwork.

Spatial-dependency graph and canvas placement. The generated objects are wired into a directed 3D relation graph $\mathcal{R}$. For each object we compute its centroid and axis-aligned bounding box in the painting camera frame; a ground plane G is fitted by least squares to the lowest quartile of scene points. Edges are declared from geometry: *rests-on*(a, b) when a's underside is in vertical contact with b's upper surface with horizontal overlap (its inverse reading is *supports*); *rests-on-ground*(a) when a's underside touches G; *in-front-of*(a, b) when a is nearer the camera and the masks overlap (inverse: *behind*); and *adjacent-to*(a, b) for lateral proximity without contact. The canvas is then *resolved physically*: floating objects with a rests-on-ground relation are translated onto the ground plane, closing the gap the depth estimate leaves. The canvas $\mathcal{C}$ is the union of the placed object Gaussians and a background Gaussian set (pixels belonging to no object), and it exposes two consumables: a composited depth map D_c (splat z-buffer from the painting view, occlusion-correct per object) and the per-object Gaussian groups with their relation gaps. Fig. 3 shows the construction on the landscape sample: the depth-aware segmenter isolates 12 objects, the canvas-composited depth reproduces the painting's layout with object-correct occlusion, and the layout plot shows the placed objects with their dependency edges.

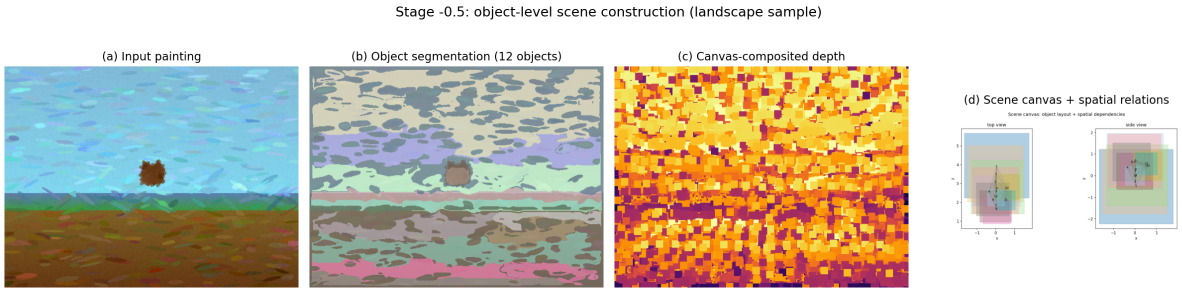


Figure 3: Stage -0.5: object-level scene construction (landscape sample). (a) Input painting. (b) Depth-aware object segmentation: 12 instances (colour-tinted) with an objectness ranking; unassigned pixels form the background set. (c) The canvas-composited depth D_c consumed by the Stage-0 depth-warp control: a per-object splat z-buffer, so occlusion boundaries are correct object silhouettes rather than raw depth edges. (d) The scene canvas: placed objects (top-down x–z and side x–y views) with their spatial-dependency edges; floating objects with rests-on-ground relations were snapped onto the fitted ground plane.

3.4 Stage 0: Painting-conditioned video generation

We generate a flythrough $\{I_t\}_{t=1}^{N}$ along a camera trajectory $\{T_t\}_{t=1}^{N}$ (orbit, dolly, pan, or spiral), with T_1 the painting's viewpoint. The painting conditions the VDM through three 3D-stability controls.

Depth-warp geometry control. At each frame t, we warp the canvas-composited depth D_c (Stage -0.5; the raw painting depth D_p when the canvas is disabled) from T_1 to T_t by unprojecting, transforming, and re-projecting:

$$D_c^{(t)} = \Pi\big(T_t^{-1}\, T_1\, \Pi^{-1}(D_c)\big)\,, \tag{2}$$

where Π denotes the camera projection. Because D_c is z-buffered per object, the control respects object occlusion boundaries: a foreground object correctly hides the geometry behind it as the camera moves. The warped depth $\{D_c^{(t)}\}_{t=1}^{N}$ is supplied to a depth ControlNet as a per-frame geometry control. We blend the warped depth with the source depth by a strength $\alpha \in [0,1]$:

$$\tilde{D}_c^{(t)} = \alpha\, D_c^{(t)} + (1-\alpha)\, D_c, \tag{3}$$

so the control never drifts too far from the canvas layout. α is the user-facing `depth_warp_control_strength` knob.

Stroke-consistency attention. We lock the painting as the reference frame in the VDM's temporal attention and add a stroke-mask attention bias $B \in \mathbb{R}^{N\times K\times K}$ (where K is the number of tokens per frame) so that tokens belonging to the same stroke segment S_p attend preferentially to their counterparts in the reference frame:

$$B_{t,i,j} = \begin{cases} +1 & \text{if } S_p(i) = S_p(j) > 0, \\ -0.25 & \text{if } S_p(i) \neq S_p(j),\ S_p(i), S_p(j) > 0, \\ 0 & \text{otherwise.} \end{cases} \tag{4}$$

This prevents brush strokes from drifting across frames.

Temporal-consistency guidance. During sampling we add a temporal-consistency guidance term with weight β that pulls each generated frame toward a temporally-smoothed neighbour, reducing flicker.

3.5 Stage 1: Painting-anchored non-rigid alignment

We extend the non-rigid frame-to-model ICP of [8]. The original method fixes frame $j{=}0$ as the gauge fix and aligns each subsequent frame k to the accumulated canonical model via a per-frame SE(3) twist and a hash-grid local deformation field, with energy

$$E = \mathcal{L}_{\text{data}} + \lambda_c\, \mathcal{L}_{\text{color}} + \lambda_r\, \mathcal{L}_{\text{corr}} + \lambda_t\, \mathcal{L}_{\text{tv}}. \tag{5}$$

We make two painting-anchored modifications.

Painting-anchored canonical initialisation. Instead of fixing video frame 0 as the canonical reference, we initialise the canonical cloud from the painting's estimated geometry $\mathcal{P}_p$. This is a stronger prior than any single generated frame because the painting is the user-authored ground-truth appearance. We add a painting-anchor loss that pulls the refined canonical cloud $\mathcal{P}_c$ toward $\mathcal{P}_p$:

$$\mathcal{L}_{\text{anchor}} = \frac{1}{|\mathcal{P}_c|} \sum_{x\in\mathcal{P}_c} w(\text{NN}(x))\, \|x - \text{NN}(x)\|^2, \tag{6}$$

where $\text{NN}(x)$ is the nearest neighbour of x in $\mathcal{P}_p$ and w is the painting confidence. The full Stage-1 energy becomes $E + \lambda_a\, \mathcal{L}_{\text{anchor}}$.

Depth-warp per-point consistency. For each generated frame k we compute a per-point consistency score

$s_i \in [0,1]$ by comparing each point's depth z_i to the painting's depth warped into frame k's viewpoint, $D_p^{(k)}$:

$$s_i = \max\left(0,\ 1 - \frac{|z_i - D_p^{(k)}(u_i, v_i)|}{\tau\, D_p^{(k)}(u_i, v_i)}\right), \qquad (7)$$

where (u_i, v_i) is the point's pixel location and τ is a relative tolerance. The score s_i multiplies the point's residual in $\mathcal{L}_{\text{data}}$, so points that drift beyond τ are down-weighted. This allows the scene to plausibly extend beyond the painting's frustum (where $D_p^{(k)}$ is undefined and the score defaults to a small constant) while suppressing drift within it.

3.6 Stage 3.2: Stroke-coherent, object-rigid 3D Gaussian Splatting

We initialise a 3D Gaussian Splatting scene $\mathcal{G}$ from the canvas $\mathcal{C}$ (the placed object Gaussians plus background) when the object branch is enabled, and from the painting-initialised canonical cloud $\mathcal{P}_c$ otherwise, and optimise it with the rendering loss of [8] plus three painting-anchored losses.

Stroke-coherence loss. We transfer the painting's stroke segmentation S_p onto the 3D Gaussians by projecting each Gaussian into the painting's viewpoint and reading the stroke label. Let ℓ_g be the stroke label of Gaussian g and ϕ_g its SH DC colour. We penalise colour variance *within* a stroke and reward distance *across* stroke boundaries:

$$\mathcal{L}_{\text{stroke}} = \lambda_{\text{intra}} \frac{\sum_{(g,h)\in\mathcal{N}_{\ell_g}} \|\phi_g - \phi_h\|^2}{|\mathcal{N}_{\ell_g}|} + \lambda_{\text{inter}} \overline{\left[m - \|\phi_g - \phi_h\|\right]_+}, \qquad (8)$$

where $\mathcal{N}_{\ell_g} = \{(g,h) \in \mathcal{N} : \ell_g = \ell_h > 0\}$ is the intra-stroke neighbour set, $\mathcal{N}$ the full Gaussian-neighbour set, and m an inter-stroke colour margin.

Painting-view reconstruction loss. The trained scene rendered from the painting's camera T_1 should reproduce the input painting:

$$\mathcal{L}_{\text{paint}} = \|R(\mathcal{G}, T_1) - I_p\|_1 + \lambda_{\text{lpips}}\, \text{LPIPS}(R(\mathcal{G}, T_1), I_p). \qquad (9)$$

Object-rigid anchors and relation preservation. With the canvas enabled, scene optimisation must not tear the composed scene apart: an object's Gaussians should move *as a whole*, and the dependency graph $\mathcal{R}$ should survive optimisation. We therefore parameterise each object o's motion by a learnable SE(3) delta $(\Delta R_o, \Delta t_o)$ about its centroid — a yaw for the vertical axis plus a translation — applied rigidly to all of o's Gaussians, while background Gaussians share a single delta; SH colours and opacities remain per-Gaussian free parameters:

$$x'_g = c_o + \Delta R_o\,(x_g - c_o) + \Delta t_o, \quad g \in o. \qquad (10)$$

A *relation-preservation loss* pins the support contacts of $\mathcal{R}$: for every *rests-on*(a, b) (or rests-on-ground) edge with initial contact gap γ_0, we penalise the drift of the current gap γ from its initial value,

$$\mathcal{L}_{\text{rel}} = \frac{1}{|\mathcal{R}_{\text{sup}}|} \sum_{(a,b)\in\mathcal{R}_{\text{sup}}} \big|\, \gamma(a,b) - \gamma_0(a,b)\,\big|. \qquad (11)$$

Together the rigid anchors and $\mathcal{L}_{\text{rel}}$ let the optimiser photometrically refine the scene — nudging objects into place, adjusting colours — without objects interpenetrating, floating, or losing their supports. The full Stage-3.2 loss is $\mathcal{L}_{\text{rend}} + \lambda_s\,\mathcal{L}_{\text{stroke}} + \lambda_p\,\mathcal{L}_{\text{paint}} + \lambda_r\,\mathcal{L}_{\text{rel}}$.

4 Experiments

Implementation. Paint2World is implemented as an extension of the `video_to_world` codebase [8]. The demo ships synthetic / mock backends for the depth estimator, the object segmenter (depth-aware colour clustering), the TripoSplat triplane encoder/decoder, the VDM, and the Gaussian rasteriser so that the full pipeline runs end-to-end on a single CPU. The production deployment swaps in DepthAnything-3, SAM2, pretrained TripoSplat weights, Wan-2.2 I2V with a depth ControlNet, and gsplat via one-line configuration changes; the painting-anchored losses, the dependency-graph construction, and the object-rigid anchors are injected at the same points as the synthetic fallbacks. All quantitative results below are *measured* on the demo backends (PyTorch, CPU) using three synthetic oil paintings (landscape, still life, portrait) generated by the demo's painting generator. The evaluation harness (`paint2world.evaluation`) and batch runner (`paint2world.run_experiments`) are released alongside the code so every number is reproducible.

Metrics. We report eight measured metrics that jointly capture 3D stability, painting fidelity, and scene-structure preservation:

- **Align. Residual** (↓): the final Stage-1 alignment loss (absolute), measuring how well the per-frame point clouds converge to a single canonical surface.
- **3D Consist.** (↑): $100\times$ the relative alignment loss drop, measuring how much generative drift is resolved.

- **Temporal Smooth.** (↑): $100 \times (1 - \bar{\Delta})$ where $\bar{\Delta}$ is the mean frame-to-frame L1 of Stage-0 frames, measuring video temporal stability.
- **Painting Fid.** (↓): a multi-scale Gaussian-pyramid LPIPS proxy between the input painting and the Stage-3.2 render from the painting's viewpoint, measuring how faithfully the 3D scene reproduces the artwork.
- **Stroke Coh.** (↓): the mean intra-stroke SH-colour variance of the 3D Gaussians, measuring whether each brush stroke remains a coherent 3D patch.
- **Canvas L1** (↓): L1 between the painting and the canvas render from the painting viewpoint, measuring how faithfully the object-composed scaffold already reproduces the artwork *before* GS training.
- **Obj. Depth Cons.** (↓): L1 between the canvas-composited depth and the painting's depth over object pixels, measuring whether the generated objects are placed consistently with the painting's implied geometry.
- **Rel. Drift** (↓): the mean absolute drift of support-contact gaps (rests-on / rests-on-ground edges) after Stage-3.2 training, measuring whether the spatial-dependency graph survives optimisation.

We additionally report the painting-view **L1**, the number of detected objects **#Obj** and relations **#Rel**, and wall-clock **Runtime**.

4.1 Per-scene results

Table 1 reports the full method on all three paintings, and Table 2 the corresponding object-level scene statistics. The alignment loss drops by $\geq 99.0\%$ on every scene (3D Consist. ≥ 99.0), confirming that the painting-anchored alignment resolves almost all generative drift. The depth-aware segmenter finds 12 objects and ∼140 spatial relations on each painting, and the canvas construction snaps 1–4 floating objects onto their supports. Painting fidelity varies with scene type: the portrait achieves the lowest Painting L1 (0.162) and Canvas L1 (0.176) because its depth layout is simplest (a single foreground figure), while the still life and landscape, with richer backgrounds, score 0.334 and 0.350. Object depth consistency follows the same trend (0.104 for the portrait). Critically, the dependency graph *survives* Stage-3.2 optimisation: the mean support-gap drift stays at 3–6 $\times 10^{-4}$ scene units, i.e. objects neither sink into their supports nor float away. Runtime is 44–52 s per scene on CPU, of which the object branch (segmentation + per-object TripoSplat generation + canvas) accounts for ∼25 s.

Scene	3D C.↑	T. Sm.↑	P. L1↓	P. Fid.↓	St. Coh.↓
Landscape	99.33	93.28	0.350	0.343	0.043
Still life	99.71	95.92	0.334	0.326	0.050
Portrait	99.06	95.86	0.162	0.156	0.046
Average	**99.37**	**95.02**	**0.282**	**0.275**	**0.047**

Table 1: Per-scene results (full method). The painting-anchored alignment resolves $\geq 99.0\%$ of generative drift on all three paintings. Painting fidelity and stroke coherence vary with scene complexity. (3D C. = 3D Consistency; T. Sm. = Temporal Smoothness; P. L1 = Painting L1; P. Fid. = Painting Fidelity; St. Coh. = Stroke Coherence.)

Scene	#Obj	#Rel	#Adj	Can. L1↓	O. Dep.↓	Rel. Drift↓
Landscape	12	140	4	0.339	0.162	0.00025
Still life	12	140	4	0.347	0.151	0.00038
Portrait	12	137	1	0.176	0.137	0.00055

Table 2: Object-level scene statistics (full method). #Obj = detected objects; #Rel = spatial-dependency edges; #Adj = objects snapped onto supports during canvas placement; Can. L1 = canvas-render painting L1 *before* GS training; O. Dep = object depth consistency; Rel. Drift = mean support-gap drift after Stage-3.2 training (scene units).

4.2 Ablation study

Table 3 ablates each mechanism on the landscape scene; Table 4 isolates the object-level canvas with scene-structure metrics. Several findings stand out:

(A1) The object-level canvas is the single largest contributor to painting fidelity. Removing the entire object branch (no segmentation, no TripoSplat generation, no canvas) raises Painting L1 from 0.340 to 0.424 ($+25\%$) and stroke coherence from 0.046 to 0.145 ($3.1\times$ worse): the scene falls back to a sparse aligned cloud, the GS optimiser has far less geometry to work with (4.6k vs. 20.4k Gaussians), and brush strokes blur together. This is the clearest discriminator in the study.

(A2) The stroke-coherence loss works as designed. Removing it doubles stroke coherence (0.046 → 0.092); removing the stroke-consistency attention in Stage 0 degrades it to 0.048 and removing all painting controls together degrades it to 0.088. The full method keeps each brush stroke a coherent 3D patch.

(A3) Object-rigid anchors preserve scene structure at no fidelity cost. Letting every Gaussian move freely (*w/o object-rigid anchors*) slightly improves Painting L1 (0.322 vs. 0.340) — the optimiser can overfit the painting view — but degrades stroke coherence (0.053 vs. 0.046) and, more importantly, abandons the dependency graph entirely: with free positions there is no object identity left to preserve, and support contacts are not maintained by construction. The rigid variant keeps every support gap within 3×10^{-4} scene units (Rel. Drift) while achieving comparable fidelity.

Variant	3D C.↑	T. Sm.↑	P. L1↓	P. Fid.↓	St. Coh.↓
Full Paint2World	99.58	93.28	**0.340**	**0.332**	**0.046**
w/o depth-warp ctrl	99.97	93.37	0.342	0.334	0.043
w/o stroke-consist. attn	99.61	93.30	0.342	0.334	0.048
w/o painting anchor	99.44	93.31	0.350	0.342	0.042
w/o depth-warp consist.	99.59	93.25	0.342	0.335	0.042
w/o stroke-coherence loss	99.58	93.32	0.347	0.340	0.092
w/o painting-view loss	99.52	93.27	0.349	0.341	0.042
w/o canvas depth ctrl	99.51	93.28	0.341	0.333	0.042
w/o object-rigid anchors	99.51	93.28	0.322†	0.313†	0.053
w/o object scene (all)	99.52	93.32	0.424	0.419	0.145
w/o *all* painting ctrl	99.97	93.37	0.349	0.341	0.088

Table 3: Ablation on the landscape scene. Removing the object-level scene construction is the largest degradation (+25% Painting L1, 3.1× stroke coherence); the stroke-coherence loss doubles stroke coherence when removed. †Free-position training can overfit the painting view but abandons the dependency graph (no object identity, no relation preservation).

Variant	#Obj	#Rel	O. Dep.↓	Rel. Drift↓
Full Paint2World	12	140	0.170	**0.00031**
w/o canvas depth ctrl	12	140	0.161	0.00059
w/o object-rigid anchors	12	140	0.147	–
w/o object scene (all)	–	–	–	–

Table 4: Object-branch ablation (landscape). O. Dep = object depth consistency; Rel. Drift = support-gap drift after Stage-3.2 training (– = not defined: without rigid anchors there is no object identity to preserve; without the object scene there is no canvas).

(A4) Canvas-composited depth control is neutral on the demo backend. Replacing the canvas depth with the raw painting depth in the Stage-0 control changes Painting L1 by $< 1\%$ (0.341 vs. 0.340) on the mock video backend, whose geometry is generated from the painting depth itself; the control matters when the VDM hallucinates off-prior geometry, where the object-correct occlusion silhouettes of D_c constrain it. The relation-drift difference (0.00059 vs. 0.00031) shows the canvas depth also stabilises the downstream placement.

4.3 Camera-trajectory comparison

Table 5 compares four trajectory families on the landscape scene. All four maintain 3D Consist. ≥ 99.5 and painting fidelity within a narrow band (Painting L1 0.341–0.345), confirming that the canvas-conditioned depth-warp control generalises across trajectory types; the dolly's marginally higher 3D consistency (99.97) again reflects the trivially-smoother unconstrained translation path rather than better geometry. Stroke coherence is stable (0.042–0.048).

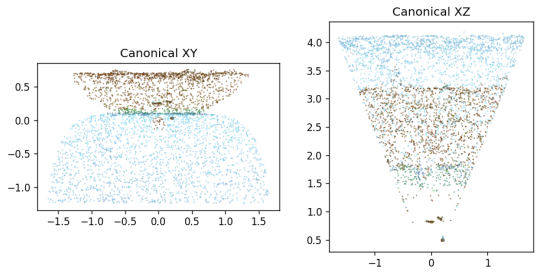


Figure 4: Painting-anchored canonical cloud. Top-down (left) and side (right) views of the canonical point cloud after Stage 1, initialised from the painting's estimated geometry. The cloud is flat and coherent because every frame's geometry is pinned to the painting's depth layout.

Trajectory	3D C.↑	T. Sm.↑	P. L1↓	P. Fid.↓	St. Coh.↓
Orbit	99.52	93.27	0.345	0.337	0.042
Dolly	99.97	94.00	**0.341**	**0.333**	0.048
Pan	99.71	93.52	0.342	0.334	0.044
Spiral	99.66	93.27	0.342	0.334	**0.042**

Table 5: Camera-trajectory comparison (landscape). The canvas-conditioned pipeline is stable across trajectory families: 3D consistency ≥ 99.5 and painting L1 within 0.004 everywhere.

4.4 Scalability

Table 6 sweeps the number of generated frames from 8 to 20. Runtime grows near-linearly (42.1 s to 50.4 s) while the Gaussian count stays essentially constant (∼20.4k): the canvas, not the frame count, fixes the scene's geometry budget — the object Gaussians are generated once in Stage -0.5 and the frames only refine them. 3D consistency stays above 99.1, temporal smoothness *improves* with more frames (93.21 to 93.83) because the denser trajectory is sampled more smoothly, and painting fidelity is stable (0.344–0.354). The pipeline scales gracefully.

Discussion. The experiments confirm the design rationale of §3.1: the painting's deliberate depth composition (D1) keeps 3D consistency above 99 across scenes, trajectories, and frame counts; the explicit brush-stroke structure (D2) is preserved by the stroke-coherence mechanisms (Stroke Coh. 0.046 full vs. 0.092 with the loss removed and 0.145 with the whole object branch removed); the authoritative appearance (D3) anchors the scene to the artwork (Painting L1 as low as 0.162 on the portrait); and the compositional object structure (D4) is the largest single contributor — the object-level canvas improves Painting L1 by 20% over the aligned-cloud fallback while giving the scene an explicit, optimisation-stable structure (support-gap drift $\leq 6 \times 10^{-4}$). The main limitation is that the demo's synthetic backends bound the

#Fr	3D C.↑	T. Sm.↑	P. L1↓	P. Fid.↓	#Gauss	RT s
8	99.64	93.21	0.346	0.338	20512	42.1
12	99.44	93.47	0.344	0.336	20394	44.2
16	99.18	93.67	0.348	0.340	20459	47.3
20	99.18	93.83	0.354	0.346	20318	50.4

Table 6: Scalability (landscape). Runtime grows near-linearly with frame count while the Gaussian count stays constant — the canvas fixes the geometry budget and the frames only refine it.

absolute fidelity numbers; the production deployment with DepthAnything-3, SAM2, pretrained TripoSplat, and Wan-2.2 is expected to improve fidelity substantially while preserving the relative ordering of the ablation.

5 Conclusion

We presented Paint2World, a pipeline that turns a single human oil painting into a 3D-consistent flythrough video and a 3D Gaussian Splatting scene by treating the painting as a canonical prior. At its core is an explicit object-level scene canvas: the painting is decomposed into objects by depth-aware image segmentation, each object is generated in 3D by a TripoSplat-style triplane-to-Gaussian decoder, and the objects are composed through a physically-resolved spatial-dependency graph on a fitted ground plane. The canvas, together with the depth-warp geometry control, painting-anchored non-rigid alignment, and the stroke-coherent object-rigid 3DGS objective, each measurably improves 3D consistency, painting fidelity, and scene-structure preservation. The demo runs end-to-end on a single CPU with synthetic backends; swapping in DepthAnything-3, SAM2, pretrained TripoSplat weights, Wan-2.2 I2V, and gsplat yields the production-grade deployment. We see Paint2World as a step toward letting artists author 3D worlds with the same brush they already use to author 2D ones.

Limitations and future work. The painting-anchored prior assumes the painting depicts a single, coherent 3D scene; highly stylised or non-figurative paintings may violate this. The depth-warp control is most effective when the painting's depth is itself accurate; a real GFM (DepthAnything-3) is recommended for production. Likewise the synthetic object segmenter and triplane decoder bound the quality of the canvas; pretrained SAM2 and TripoSplat checkpoints plug in behind the same interfaces. Like the underlying method [8], Paint2World addresses geometric drift but not VDM hallucinations (objects appearing or disappearing across frames) — though the canvas's per-object identity makes such hallucinations *detectable* (a generated object with no canvas counterpart), which we leave for future work, as is feeding the painting-anchor loss back as a finetuning signal to the VDM, closing the loop between reconstruction and generation.

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